HTML Standards

There are, of course, dozens of other HTML tags, and there are also some rules about how they should be structured. HTML that conforms to these rules is said to be *valid*, and it's generally a good idea to make sure your PHP is producing valid HTML. Malformed HTML (missing closing tags, broken hierarchy, etc.) is *invalid* and can cause issues for clients (web browsers can cope with invalid HTML to an extent but might behave unpredictably). As we discussed earlier, HTTP (and a plethora of other internet technologies) is defined by IETF standards documents (RFCs). HTML is, of course, also governed by standards, which have evolved greatly since its beginnings in the 1990s. Modern HTML should follow the HTML5 standard, which is maintained by the Web Hypertext Application Technology Working Group (WHATWG). The HTML5 "Living Standard" can be found at `https://html.spec.whatwg.org/dev/` and is quite a read, so let's ease you in gently…

All minimally valid HTML documents have the following basic structure:

```
<!DOCTYPE html>
<html>
  <head>
    <title>My page.</title>
  </head>
  <body></body>
</html>
```

The first line of the example above is the *doctype declaration* and is "required preamble" according to the HTML5 specification. This line gives fundamental instructions to the client about the required rendering mode. It is case-insensitive but is conventionally given in the form shown, with capitalized `DOCTYPE`, one or more whitespaces, and lowercase `html`. The doctype line does not form part of the HTML document itself: that begins with the first start (opening) tag.

Our first start tag is `<html>`; it is the root-level element in our document (also known as "document element" because its parent is the document itself). It is possible to omit this tag and the corresponding `</html>` closing tag if there are no comments (an HTML comment has the syntax `<!-- a comment -->`) before the next start tag (`<head>` in our case) or after the final closing tag. The following is also considered valid HTML5…